

Ayurvedic Based Disease Prediction & Personalized Medicine & Diet Recommendation System with Intelligent QA Support

25-26J-305



Introduction to the Research Problem

- People in rural areas often cannot access qualified Ayurvedic doctors due to long distances and high costs.
- Many rely on self-medication or generic online advice, which can be unsafe.
- 91.1% of Sri Lankans prefer Ayurvedic remedies at their fingertips (Survey, n=235).

Affected Parties:

- **Patients:** Delayed diagnosis, inconsistent treatments, lack of personalized care
- **Rural Communities:** Limited access to specialized Ayurvedic knowledge
- **Traditional Practitioners:** Need for decision support tools

Importance of Solving This Problem

Immediate Healthcare Impact

- Prevent unsafe self-medication affecting millions
- Reduce travelling cost and
- Bridge urban-rural healthcare divide

Long-term Societal Benefits

- **Preserve cultural heritage** - digitize traditional knowledge
- **Democratize healthcare access** across all provinces

Introduction to the Research Problem

Social Impact

- **Safety Enhancement:** Replace dangerous self-medication with AI-guided recommendations
- **Cultural Preservation:** Digitize knowledge from practitioners
- **Health Equity:** Equal access regardless of geographic location

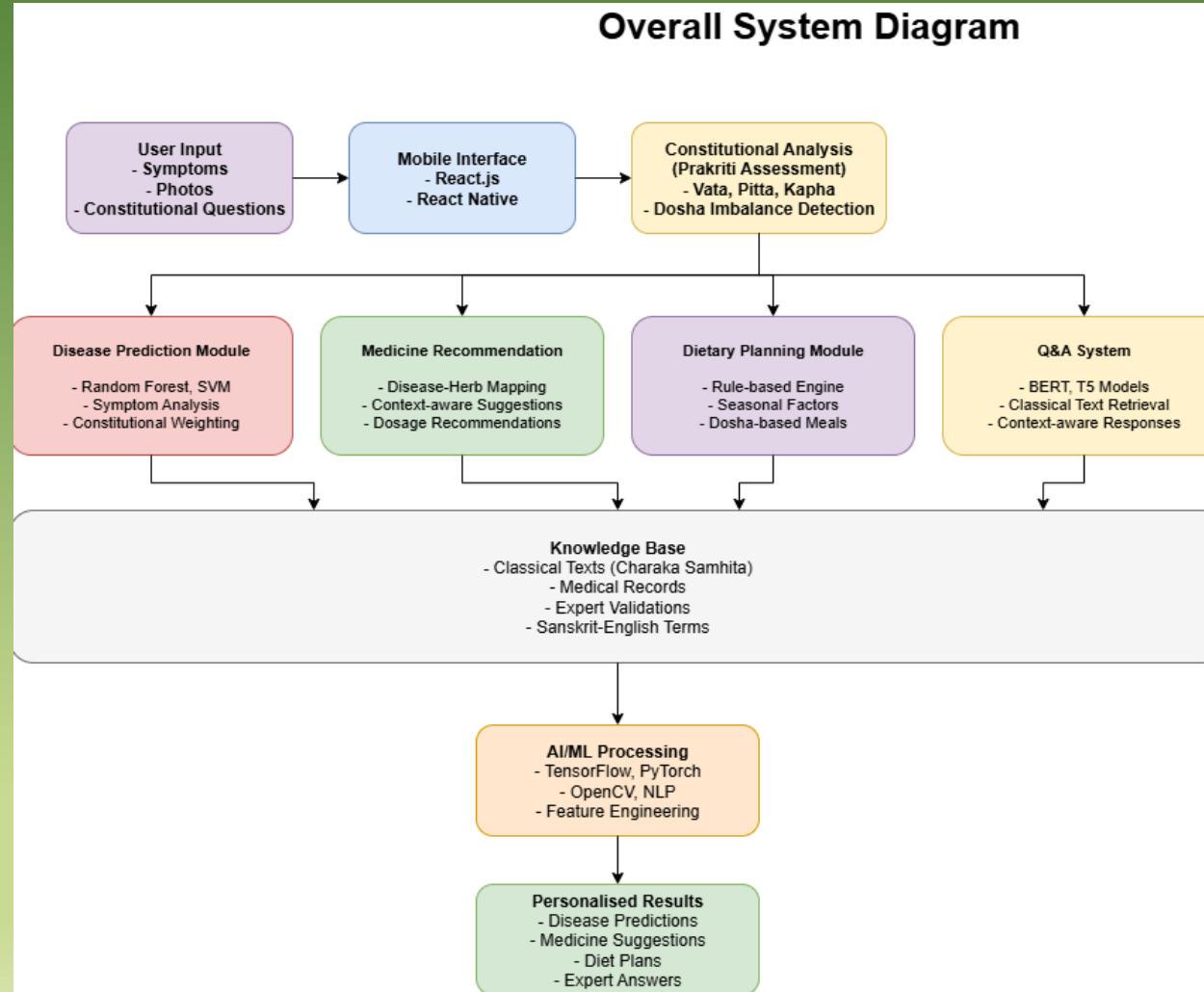
Research Impact

- Explores the application of machine learning in traditional medicine systems
- Evidence-Based Medicine: Enable clinical validation of traditional treatments

Constraints and Limitations

- Data Availability
- Accuracy
- Scope
- Cultural Acceptance

Introduction to the Research Problem





IT22344274 | J. P. ROCHE

Bsc (Hons) in Information Technology Specializing in Information Technology

AYUVEDIC MEDICINE RECOMMENDATION MODULE

- Lack of Personalized Symptom-Based Herb Recommendations
- Limited Integration of AI/ML for Herb Prediction
- Absence of Comprehensive Knowledge Base

Proposed Solution :

- Creation of a Structured Ayurvedic Knowledge Base
- Advanced NLP with Named Entity Recognition to automatically extract and categorize medical information from Text
- Personalized Recommendation that predicts the most suitable Herb for a Disease Based on user factors
- Provides a **user-friendly interface** for easy access

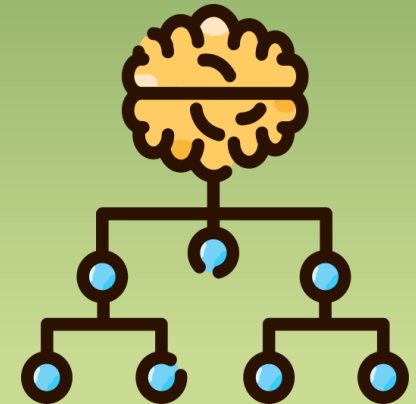
Ayurvedic Medicine Recommendation Module

Key Domains of specialization-based Knowledge

- **Machine Learning** → Classification algorithms to predict most suitable Herb(S)
- **Natural Language Processing (NLP)** → NER to extract entities from Text
- **Healthcare & Ayurveda Domain Knowledge** → mapping symptoms to herbs, traditional medical practices.
- **Software Engineering** → system design, user interface, integration of modules.

specific Latest Technology/Technologies

- **ML:** Scikit-learn, TensorFlow, PyTorch.
- **NER:** spaCY
- **Recommender Systems:** train classification model (random forest, SVM, decision trees)
- **Databases:** storing herb-symptom mappings.



Ayurvedic Medicine Recommendation Module

Evaluate / Prove the Success

Metrics:

- Classification accuracy, precision, recall, F1-score.
- Ayurvedic Practitioner Review

Testing:

- Train/test split on dataset

Expected Accuracy:

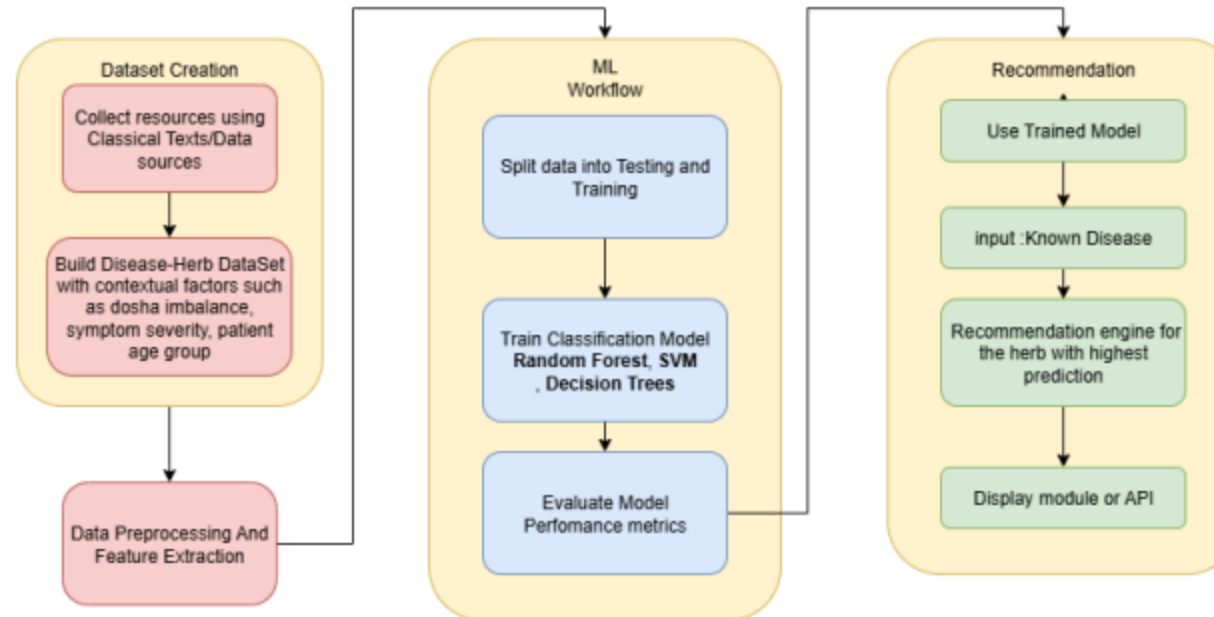
- Aim for 70–90% classification accuracy

Data Availability

- Classical Texts
- Hospital Records

AYUVEDIC MEDICINE RECOMMENDATION MODULE

Component Diagram





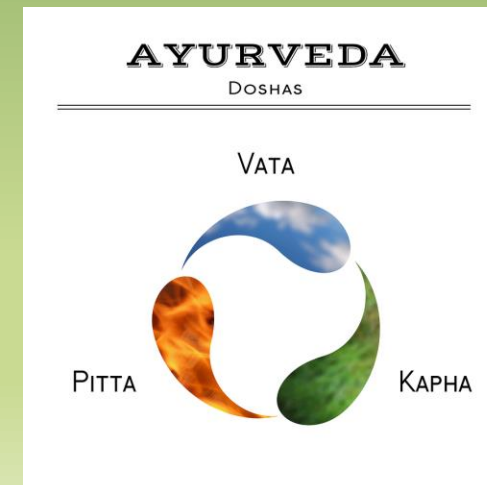
IT22905918 | S. I. A. PERERA

Bsc (Hons) in Information Technology Specializing in Information Technology

CONSTITUTIONAL- AWARE AYURVEDIC DISEASE PREDICTION MODULE

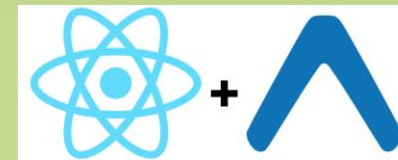
- Current Systems Limitation:** Existing disease prediction systems focus only on symptom-disease mapping without considering individual constitutional types (Prakriti)
- Missing Component:** No computational model exists for quantifying Tri-Dosha (Vata, Pitta, Kapha) imbalances in machine learning-based disease prediction
- Clinical Challenge:** Identical symptoms may indicate different conditions based on patient constitution, making generic predictions inadequate

- First Constitutional- Aware ML System
- Tri-Dosha Quantification Model
- NER for Medical Text



CONSTITUTIONAL- Aware Ayurvedic Disease Prediction MODULE

- **Machine Learning:** Implementation of classification algorithms (Random Forest, SVM, Neural Networks) for multi-label disease prediction
- **Deep Learning:** Neural network architectures for complex symptom pattern recognition and constitutional analysis
- **Natural Language Processing:** Named Entity Recognition (NER) models for Sanskrit medical terminology extraction from classical texts
- **Feature Engineering:** Novel constitutional-based feature creation and 3-level symptom severity scoring mechanisms
- **Model Evaluation:** Cross-validation, performance metrics, and comparative analysis methodologies

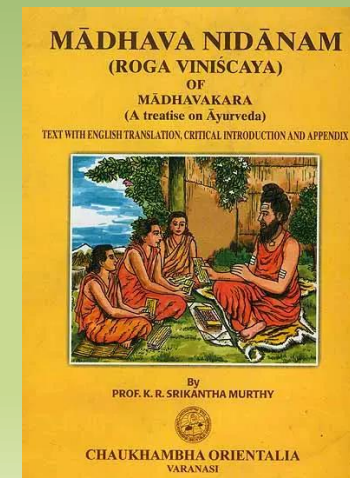
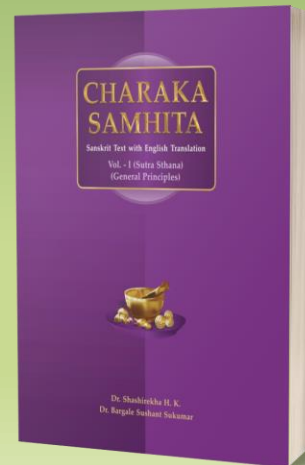


CONSTITUTIONAL- Aware Ayurvedic Disease Prediction MODULE

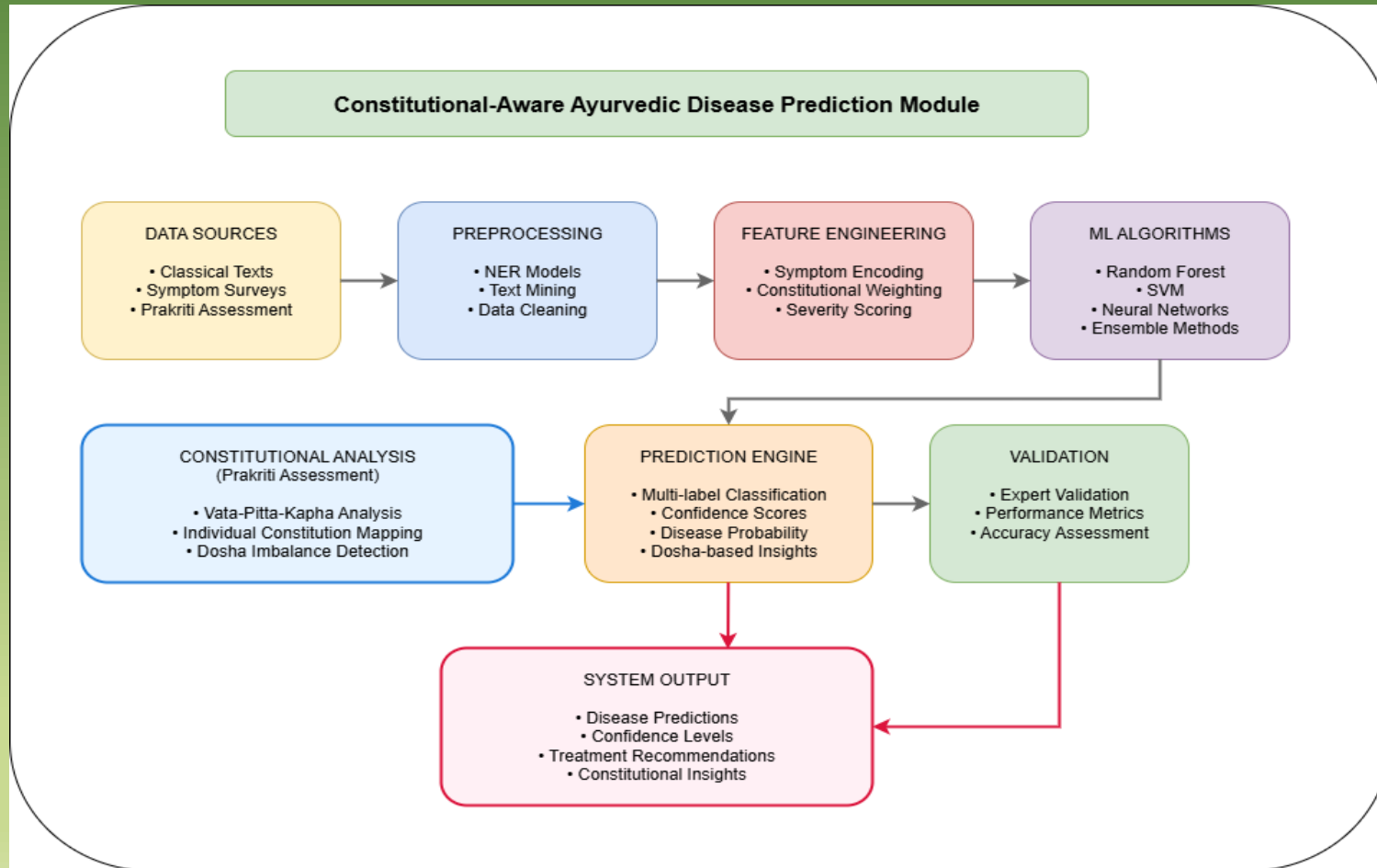
Validation & Success Metrics

- **Expert Validation:** Consultation with 2-3 qualified Ayurvedic practitioners for model accuracy verification.
- **Performance Metrics:** F1-score, precision, recall, and accuracy measurements for disease classification.
- **Comparative Analysis:** Performance comparison with/without constitutional features to demonstrate improvement.
- **Expected Accuracy:** Target 85%+ accuracy with constitutional features & 70% without.

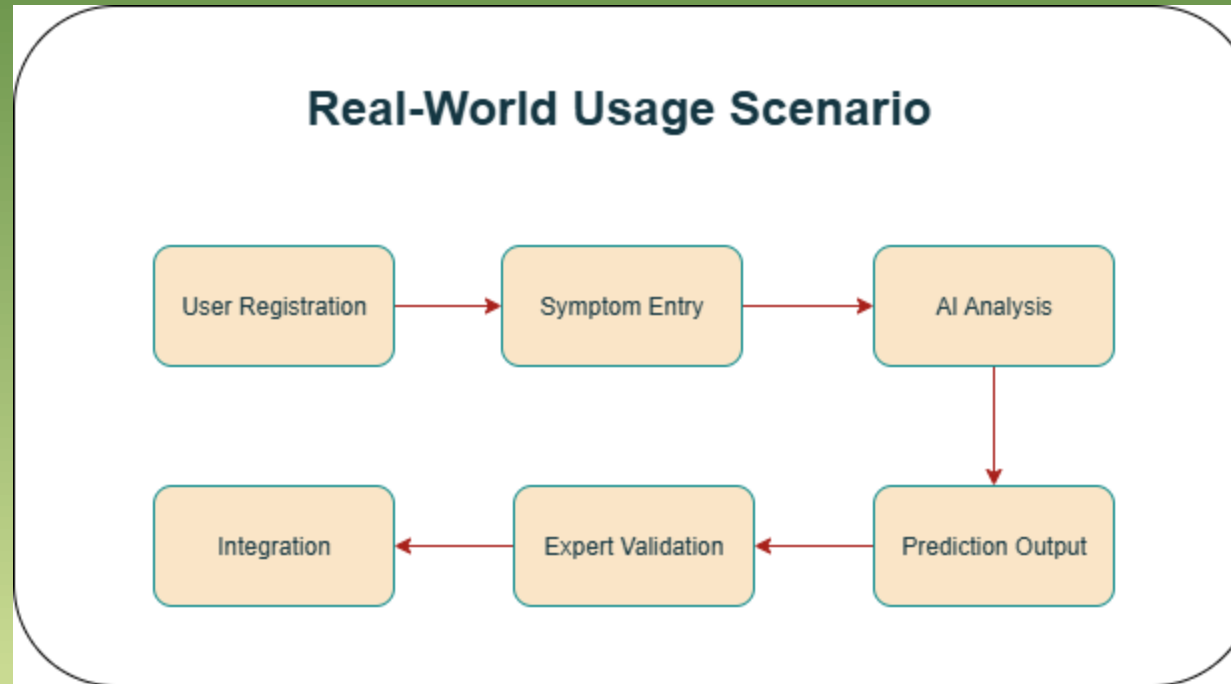
Data Availability



Constitutional- Aware Ayurvedic Disease Prediction Module



Constitutional- Aware Ayurvedic Disease Prediction Module





IT22897176 | K.P.M.R.A FERNANDO

Bsc (Hons) in Information Technology Specializing in Information Technology

INTELLIGENT QUESTION ANSWERING (QA) OVER AYURVEDIC MEDICINE / ADVICE MODULE

Knowledge Gap or Problem Definition

- Existing Ayurvedic QA tools are **English-only, rule-based**.
- No **multilingual (Sinhala/Singlish)** or **personalized (dosha-based)** support.

Shortcomings of Existing Systems

- Cannot handle open-ended or code-mixed queries.
- Generic answers, no **sutra references** for trust.
- Lack of structured datasets → low accuracy.

Proposed Solution

- **Multilingual, constitution-aware QA system.**
- Uses **RAG + Ayurvedic NER + sutra-linked explainability.**
- Provides **personalized, reliable answers** grounded in authentic texts.

INTELLIGENT QUESTION ANSWERING (QA) OVER AYURVEDIC MEDICINE / ADVICE MODULE

Domains Used

- **NLP** → multilingual embeddings, transliteration.
- **Machine Learning/Deep Learning** → transformers.
- **Information Retrieval** → FAISS vector search.

Latest Technologies

- **Retrieval-Augmented Generation (RAG).**
- **Transformer models (mT5, IndicBERT).**
- **Custom Ayurvedic NER** (doshas, herbs, sutras).

INTELLIGENT QUESTION ANSWERING (QA) OVER AYURVEDIC MEDICINE / ADVICE MODULE

Evaluation / Success Proof

- Target: >85% F1, >70 BLEU, >90% Recall@k.
- Practitioner validation ensures quality.

Data Availability

- Classical texts: Charaka Samhita, Madhava Nidana, Ayurvedic Hospital Records.
- Practitioner-reviewed QA dataset .

Expected Performance:

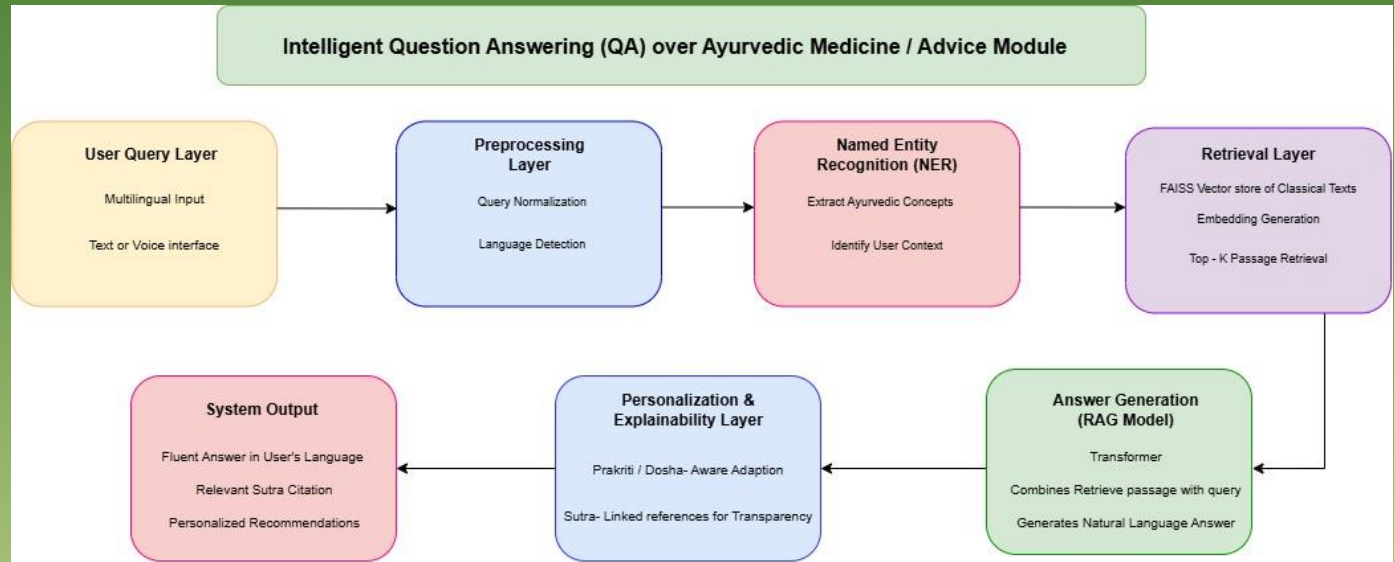
- High F1-score for NER and retrieval
- BLEU >70 for QA generation
- Reliable practitioner-reviewed outputs

Ethical Clearance

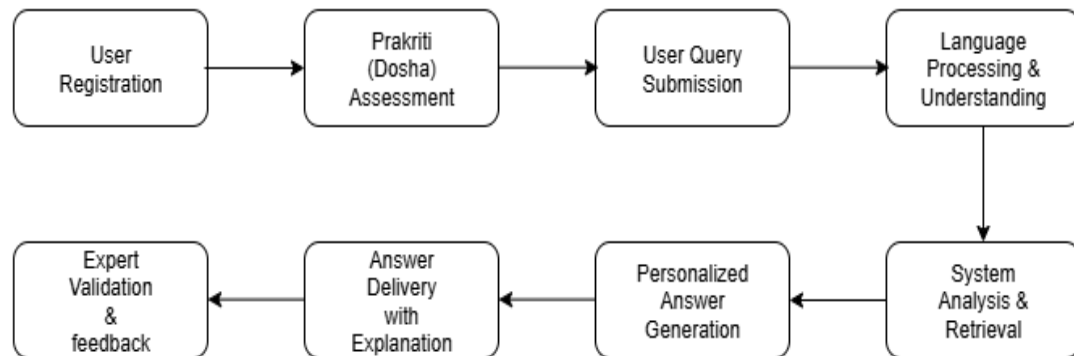
- Use **open-access digitized texts**.
- **All annotations and QA pairs reviewed by Ayurvedic practitioners** to ensure accuracy.
- Compliance with cultural sensitivity and respect for traditional knowledge.

INTELLIGENT QUESTION ANSWERING (QA) OVER AYURVEDIC MEDICINE / ADVICE MODULE

System Overview of the Component



Real-World Use





IT22899910 | W. A. N. M. DIAS

Bsc (Hons) in Information Technology Specializing in Information Technology

AYURVEDIC DIETARY RECOMMENDATION MODULE

Research Gap

- Few approaches consider adaptive personalization (digestion, mood, sleep, season, age).
- Limited integration of disease-specific dietary rules (Pathya/Apathya) into computational systems.
- Lack of practitioner-validated datasets.

Shortcomings of Existing Systems

- Existing Ayurvedic dietary systems are oversimplified, focusing only on static dosha–food suitability and ignoring contextual factors such as digestion, mood, sleep, age, and season.
- There is a lack of practitioner-validated, disease-specific computational models, limiting the clinical reliability and personalization of dietary recommendations.

Proposed Solution

- Rule-based logic from classical Ayurvedic texts (Rasa, Guna, Virya, Vipaka).
- Machine learning adaptation (KNN, Random Forest, SVM) to refine recommendations.
- Personalization loop using real-time user feedback (digestion, mood, sleep).
- Practitioner-verified validation for food-dosha-disease mappings.

AYURVEDIC DIETARY RECOMMENDATION MODULE

Ayurvedic Science & Nutrition:

Dosha theory (Vata, Pitta, Kapha), Food attributes (*Rasa, Guna, Virya, Vipaka*), Pathya–Apathya (disease-specific diets), Seasonal and contextual dietary principles

Machine Learning:

KNN, Random Forest, SVM for implementation

Natural Language Processing (NLP):

NER for extraction + Rule-based post-processing to ensure Ayurvedic authenticity.

Feature Engineering:

Creating composite features (dosha impact, pathya/apathya classification).

System Design:

User-centered interface design

Latest Technology/Technologies in that Domain are utilized

Domain	Technologies/Tools
Machine Learning (ML)	Scikit-learn, TensorFlow, PyTorch
NER (Natural Language Processing)	spaCy, BERT/BioBERT
Recommender Systems	Hybrid Rule-Based + ML (KNN, Random Forest, SVM)
Databases	MongoDB, PostgreSQL, CSV/JSON

AYURVEDIC DIETARY RECOMMENDATION MODULE

Validation & Success Metrics

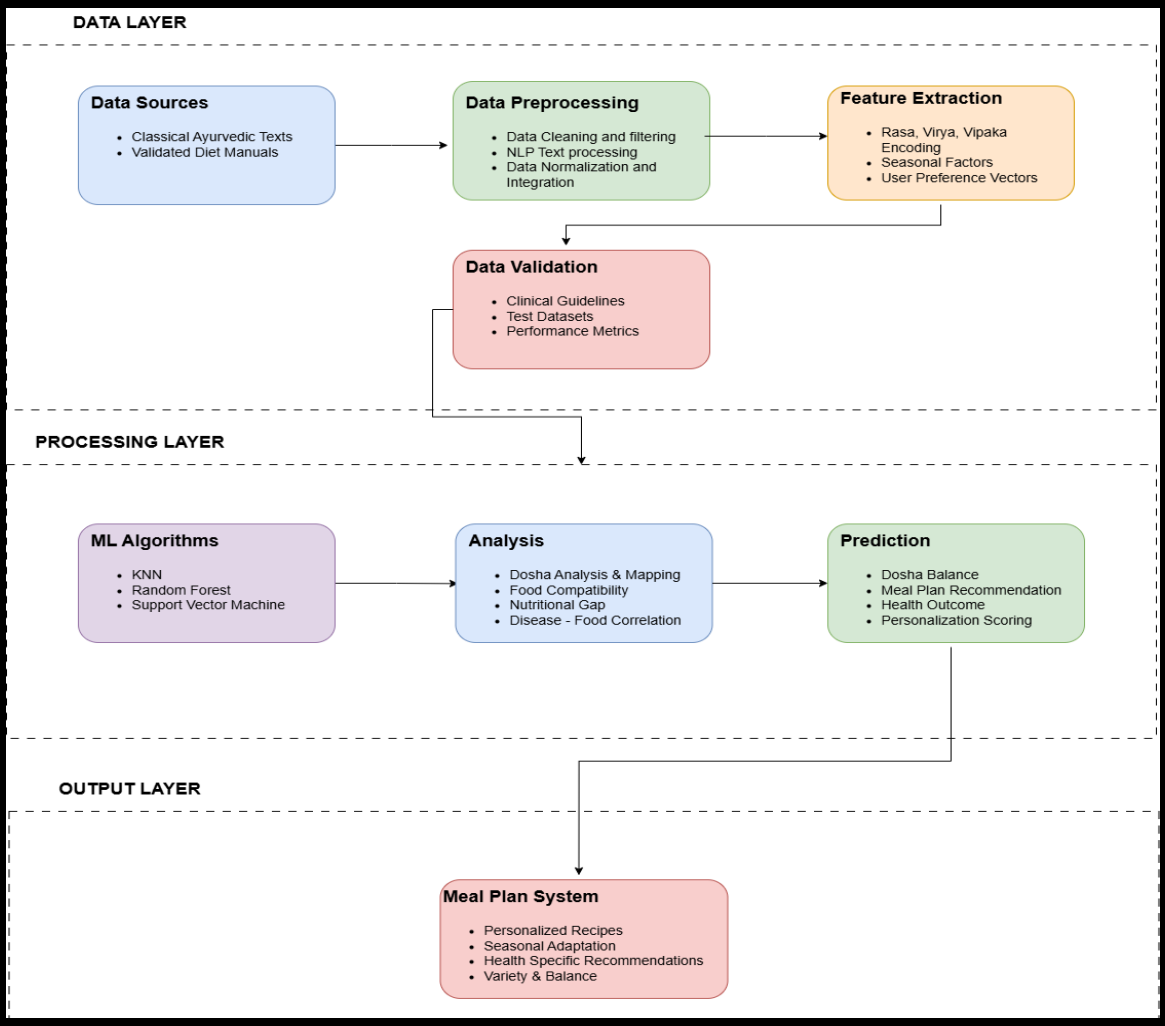
Metric	Validation Method	Target/Goal
Expert Validation	Practitioner review of food-dosha-disease mappings	Ensure authenticity
Performance Metrics	Accuracy, Precision, Recall, F1-score	≥ 85% accuracy
Comparative Analysis	Benchmark vs. existing Ayurvedic systems	Show measurable improvement
User Satisfaction	Feedback surveys (Likert scale)	≥ 80% satisfaction
Adaptability	Test with digestion, mood, sleep, seasonal inputs	Dynamic adjustments
Scalability	Stress testing with large datasets	Maintain performance
Reliability	System uptime monitoring	≥ 95% uptime

Data Availability

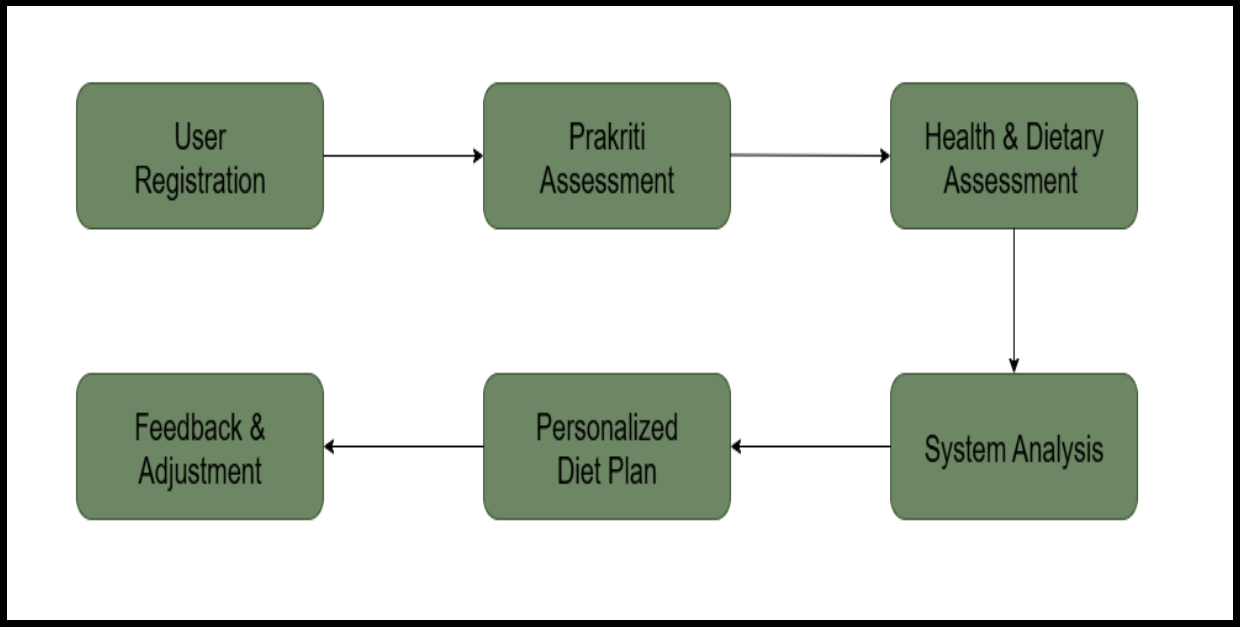
- Ayurvedic Texts and Literature (Classical Ayurvedic texts such as the *Sushruta Samhita*, *Ashtanga Hridaya*, and *Bhavaprakasha Nighantu*)
- Ayurvedic National Hospital, Sri Lanka (patient dietary records)
- Ayurvedic Practitioners

AYURVEDIC DIETARY RECOMMENDATION MODULE

System Overview



Real World Scenario



REFERENCES

- [1] H. Sharma, H. M. Chandola, G. Singh, and G. Basisht, "Health and Food in Ayurveda," Journal of Alternative and Complementary Medicine, vol. 13, no. 10, pp. 1075–1080, 2007.
- [2] B. Rathi, "An Ayurvedic Approach to Pathya: A Study on Disease-Specific Food Classification," International Journal of Ayurveda and Pharma Research, vol. 8, no. 2, pp. 45–50, 2020.
- [3] A. Ismayilova, C. Zhang, and Y. Luo, "Ayurveda-Based Recipe Recommendation System," Report, 2024. [Online]. Available: [ResearchGate / preprint or as per your source]
- [4] L. B. Rananavare and S. Chitnis, "VrikshBot: AI-Powered Conversational Guide to Vrikshayurveda," International Conference on Innovative Trends in Information Technology (ICITIIT), 2025.
- [5] V. Serrao, A. K. M, and Varsha, "AyurMate: Blending Ayurveda and Artificial Intelligence for Personalized Wellness," International Conference on Artificial Intelligence and Data Engineering (AIDE), 2025.
- [6] V. A and M. N, "AI-Driven Ayurvedic Chatbot: Revolutionizing Formulation Selection and Knowledge Accessibility in Traditional Medicine through the Integration of LENET and Stacked Autoencoders," 10th International Conference on Advanced Computing and Communication Systems (ICACCS), 2024.
- [7] A. Silva, B. Rathnayake, and C. Jayasinghe, "Human Skin Diseases Identification and Treatment Suggestion by Sri Lankan Ayurveda Medicine Using Machine Learning," in Proc. IEEE Int. Conf. Information Technology, 2024, pp. 1-6.

Commercialization

Ability of Commercialization

- High Demand Validation
- Digital Health Adoption: comfortable with health apps
- Lacks Ayurvedic AI system existence for Sri Lanka
- Cultural Alignment: Builds on existing traditional medicine usage

Investment/cost (Labor + Material)

Labor: Development (3–4 months), data collection & preprocessing, testing.

Material: Hosting fees, database subscription, software tools (Python, ML libraries).

Recovery:

- Subscription model for app users (e.g., monthly fee)
- One-time purchase for premium features
- Online consultation or personalized treatment reports

Target Market & Customer Profile:

Health-conscious individuals, Ayurveda enthusiasts, people seeking personalized herbal treatment.

- **Customer Profile:**
 - Age: 25–45
 - Urban areas, tech-savvy
 - Interested in natural remedies, willing to pay for personalized guidance

Value Proposition:

- Provides a Remote access app that predicts the disease ,recommenda medicine ,diet and a intelligent QA support
- Doesn't have to travel far saves both time and money